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Title

**Assessment of Alzheimer's Disease
Progression Using Comorbidity-Aware
Meta-Bayesian Framework with
Resting-State fMRI**

REVIEW REPORT – II

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Abstract

Background: Early detection of Alzheimer’s disease (AD) is critical to managing the growing global dementia burden. Resting-state functional magnetic resonance imaging (rs-fMRI) offers crucial insights into early functional brain network disruptions before irreversible structural damage occurs. However, existing computational approaches face major limitations: they often analyze functional biomarkers independently, omit patient comorbidity profiles, ignore the continuous nature of biological progression, and lack diagnostic uncertainty quantification.

Objective: We propose a novel framework, the Comorbidity-Aware Meta-Bayesian Multi-Attention Transformer, to learn a shared, biomarker-invariant AD representation while explicitly integrating clinical covariates and quantifying prediction uncertainty.

Methods: Resting-state fMRI scans were drawn from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) repository. The data foundation was built first: 175 acquisitions were audited, 29 missing scans were recovered from raw DICOM and 2 were excluded, and 165 acquisitions from 127 subjects were preprocessed with head-motion correction, direct normalization to the MNI template, band-limited temporal filtering and a 27-regressor nuisance model. Slice-timing correction was not possible because the metadata are not recoverable, and no T1-weighted image exists for any subject, so no anatomical co-registration was used. Four complementary functional biomarkers were then extracted—Amplitude of Low-Frequency Fluctuations (ALFF), Regional Homogeneity (ReHo), Degree Centrality (DC), and Functional Connectivity (FC)—with the brain parcellated into 246 regions via the Brainnetome Atlas, and each was normalized within subject. The proposed model will use a Multi-Attention Transformer (incorporating global, local, and network attention) to learn critical region embeddings, and a meta-learning module to distill shared, biomarker-invariant disease signatures. These imaging representations will be fused with patient-specific clinical metadata (age, APOE ϵ 4 genotype, MMSE, CDR-SB, Geriatric Depression Scale scores, and cardiovascular and endocrine medical history). A Bayesian ordinal classifier will then assign each subject a position along six sequentially ordered cognitive stages spanning the healthy-to-diagnosed continuum—CN, SMC, EMCI, MCI, LMCI, and AD—together with a calibrated confidence score for each prediction.

Current Development and Evaluation Framework: Modules 1 to 3 of the proposed system (data, preprocessing and feature extraction) are complete, one milestone ahead of the Review I timeline. The modelling stages (Modules 4 to 6) have not been started, and no performance result is reported. Every derived feature was verified independently, and degree centrality recomputed from the stored connectivity

matrices agreed exactly for all 165 acquisitions. Four findings shaped the design. FC strength, the per-region summary of FC, is exactly $DC/245$, so all four biomarkers are kept as a 246×3 regional matrix plus the full 246×246 connectivity matrix. The three regional biomarkers share only 6 to 11% of their variance across regions, so each carries largely different information. Stage-wise group-mean heatmaps over 127 subjects show no detectable univariate progression signal (1, 0 and 1 of 246 regions monotonic, against about 0.7 expected by chance), which motivates a multivariate model. The features are reproducible between sessions: with no stage label, the most similar other acquisition belongs to the same subject for 91% of the 69 acquisitions of subjects scanned more than once (mALFF), although residual head motion and an uneven spread of sites across stages remain in the features. The model will be judged on the standard classification metric suite (accuracy, precision, recall, F1-score, ROC-AUC) alongside two model-specific checks: how well calibrated the Bayesian uncertainty estimates are, and how meaningful the learned attention maps prove to be, under subject-grouped cross-validation and against single-modality baselines.

Significance and Conclusion: This work introduces a novel architectural formulation designed to advance transparent modeling of Alzheimer’s disease progression. By uniquely unifying cross-biomarker transferability, multi-attention networks, patient comorbidity profiles, and probabilistic classification, this framework ultimately aims to deliver a robust, highly explainable clinical decision support system.

Keywords: Alzheimer’s Disease Progression, rs-fMRI Biomarkers, Multi-Attention Transformer, Meta-Learning, Bayesian Ordinal Classification, Comorbidity Fusion

1. Introduction

Alzheimer’s disease is a progressive neurodegenerative disorder and the leading cause of dementia worldwide. Following the staging scheme fixed at Review I, its clinical course is divided into six ordered stages: cognitively normal (CN), subjective memory concern (SMC), early mild cognitive impairment (EMCI), mild cognitive impairment (MCI), late mild cognitive impairment (LMCI) and Alzheimer’s disease (AD). The boundaries between these intermediate stages are diagnostically important, because therapeutic and care decisions differ substantially between them, yet they are difficult to resolve from cognitive testing alone. Distinguishing them reliably from non-invasive imaging is therefore of direct clinical value.

Resting-state functional MRI measures spontaneous low-frequency fluctuations in the blood-oxygen-level-dependent (BOLD) signal while the subject is at rest. Because functional disruption in AD is believed to precede measurable structural atrophy, rs-fMRI derived measures are attractive candidate markers of early progression. Several complementary families of rs-fMRI measure exist. Amplitude of low-frequency fluctuation (ALFF) quantifies the local strength of spontaneous activity [5]; regional homogeneity (ReHo) quantifies the local temporal coherence of neighbouring voxels [4]; functional connectivity (FC) quantifies statistical dependence between distant regions; and degree centrality (DC) summarises how strongly a region participates in the whole-brain network [7]. Each captures a different aspect of the same underlying physiology.

This diversity is both an opportunity and a difficulty. Individual biomarkers are known to be noisy and to vary with acquisition protocol, so a model trained on one alone tends to generalise poorly. The central premise of this project is that a representation shared across several biomarkers — one that is invariant to which biomarker it was derived from — should be more stable than any single measure. A further complication is that AD rarely occurs in isolation: age, APOE4 genotype, depression, hypertension and diabetes all modulate both brain function and cognitive decline, and a model that ignores them attributes comorbidity-driven variation to disease stage. Finally, because staging decisions are clinical, a point prediction is insufficient; a calibrated measure of uncertainty is required for a prediction to be actionable.

The significance of the present stage of work lies in the data foundation. Every downstream modelling claim rests on the correctness of the preprocessing and biomarker extraction beneath it, and on an honest account of what the source data can and cannot support. Accordingly, this report emphasises verified, reproducible processing with explicit documentation of every point at which the dataset was found

to be limiting.

1.1 Objectives

The objectives are those fixed at Review I.

1. To identify functional brain biomarkers and comorbidity patterns associated with Alzheimer’s disease progression using resting-state fMRI.
2. To develop a comorbidity-aware feature representation framework for robust characterization of Alzheimer’s disease progression.
3. To design a Meta-Bayesian learning framework for accurate and reliable Alzheimer’s disease progression assessment.
4. To incorporate uncertainty estimation for improving the confidence and robustness of disease stage prediction.
5. To validate the proposed framework using benchmark datasets through performance, statistical, and clinical evaluation.

Table 1.1 states honestly how far each objective has been met. The work reported here is the data foundation beneath objectives 1 and 2.

Table 1.1: Progress against the Review I objectives.

Obj.	Status	Evidence
1	Partly met	ALFF, ReHo, FC and DC extracted and verified for 165 acquisitions (127 subjects). Their association with progression is not demonstrated (Section 6.4), and comorbidity patterns have not yet been analysed; the clinical table is cleaned, with covariates for 107 of the 127 subjects (Section 4.5.2).
2	Partly met	The imaging half is built from all four biomarkers: ALFF, ReHo and DC as a 246×3 regional matrix and FC as a 246×246 Fisher- z matrix per acquisition. The clinical table for the comorbidity-aware half is cleaned (covariates for 107 of 127 subjects), but the fusion has not been built.
3	Not started	No Meta-Bayesian model exists.
4	Not started	No uncertainty estimate exists.
5	Not started	Only data-level validation (independent recomputation of the biomarkers) has been done. No model has been validated; no performance, statistical or clinical evaluation exists.

1.2 Position Against the Review I Timeline

The Review I timeline (August 2026 to March 2027, Figure 1.1) scheduled *Module 2: Preprocessing & Validation* for September–October (Review II) and *Module 3: Feature Extraction — Brainnetome 246* for November (Review III). Both are complete at Review II, so on the data side the project stands a full milestone ahead of its own Review I timeline (Table 1.2).

Table 1.2: Review I timeline versus actual progress at Review II.

Review I activity	Planned	Status at Review II
Literature survey, problem formulation & design (Review 1)	Aug	Complete.
Module 2: Preprocessing & validation (Review 2)	Sep–Oct	Complete. 165 of 173 eligible acquisitions preprocessed (outputs dated 20–22 September 2026); 8 rejected for a 6.02s TR; per-acquisition QC and independent verification done.
Module 3: Feature extraction — Brainnetome 246 (Review 3)	Nov	Complete, one review early. 165/165 acquisitions parcellated with 246/246 coverage; ALFF, ReHo, FC and DC generated and audited on 23 September 2026; within-subject normalisation finished on 5 October 2026.
Modules 4 & 5: Multi-Attention Transformer, Meta-Learning & 6-class classifier	Dec–Jan	Not started. No model has been trained; no accuracy, AUC or other performance figure exists.
Testing and evaluation	Feb	Not started.
Documentation; final submission	Feb–Mar	Not yet due.

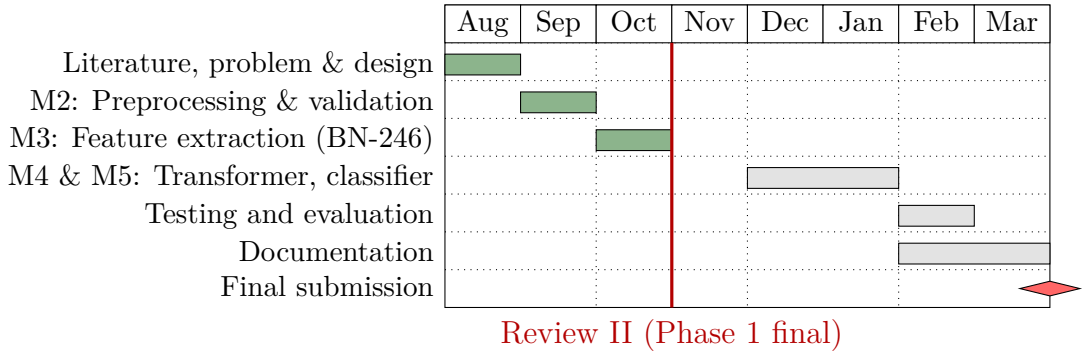


Figure 1.1: Review I timeline (bar positions) against progress at Review II. Dark bars are complete, light bars are not started. The vertical line marks Review II (October); Module 3, planned for November, was complete by early October.

The lead should be read narrowly: it applies to the data foundation only.

Module 1 is complete: the imaging is audited and the clinical table is cleaned, although covariates other than APOE $\epsilon 4$ exist for 107 of the 127 subjects (none for the LMCI group), and the cross-validation folds are not yet built (Section 6.5).

1.3 Literature Update Since Review I

Review I identified four research gaps: (G1) functional biomarkers analysed independently, (G2) clinical comorbidity profiles omitted, (G3) progression treated as a static classification, and (G4) no calibrated diagnostic uncertainty. A search for recent work since Review I found three 2026 papers that touch these gaps (Table 1.3). It was a targeted search, not a systematic review, and two of the three are arXiv preprints that have not been peer reviewed. Only what their abstracts state is reported here.

Table 1.3: Recent work against the Review I gaps. “Not verified” means the abstract does not say and the full text was not accessible.

Work	Input	Ordinal staging	Uncertainty; gaps touched
Hariharan, Gowda and Anala [17] (arXiv, 2026)	Longitudinal ADNI visits; forecasts diagnosis, CDR-SB, MMSE and hippocampal volume	CORAL ordinal head over three classes (CN, MCI, dementia)	Aleatoric (mixture variance) and epistemic (five-member bootstrap ensemble) uncertainty; G3, G4
Rauchmann et al. [18] (arXiv, 2026)	T1-weighted MRI with demographic and genetic variables; ADNI, AIBL, NIFD; subject-level splits	Ordinal regression with attention; QWK 0.549 against 0.477 for the non-ordinal model; number of stages not verified	Uncertainty not mentioned in the abstract; G2, G3
Alsalem et al. [19] (<i>Diagnostics</i> , 2026)	MRI images	Progression-aware ordinal objective; multiclass staging; number of classes not verified	Explainability reported; uncertainty not verified; G3

Effect on the gaps. G2, G3 and G4 are each partly addressed by at least one of these works, so none of them is novel on its own, and this report does not claim otherwise. G1, learning a biomarker-invariant representation across several rs-fMRI biomarkers by meta-learning, was not addressed by any of the three works checked, although this search cannot rule out other work. What distinguishes the present project is the combination: functional rs-fMRI biomarkers (ALFF, ReHo, DC, FC) as the imaging input rather than structural MRI or clinical scores; six ordered stages, where the one paper whose class count was verified uses three; comorbidity fusion; and Bayesian ordinal uncertainty. A further difference is imposed by the data: the cohort has no T1w image, so the structural-MRI route those papers rely on is not available here.

2. First Review - Queries and Answers

The panel’s Review I comments, with the response to each, are below. Each response points to the section of this report that gives the detail.

- **Comment 1** — *Need more clarity on the dataset (biomarkers).*

Response — The data are ADNI resting-state fMRI acquisitions across the six ordered stages. Of 175 raw acquisitions, 2 were excluded for documented reasons, 173 were eligible and 165 were preprocessed, from 127 distinct subjects (Table 6.1). Four biomarkers are extracted per acquisition on the 246 Brainnetome regions: ALFF, ReHo, DC and FC, defined in Table 5.2. After within-subject normalisation they enter the model as a 246×3 regional matrix (mALFF, mReHo, DC_z) and a 246×246 Fisher- z FC matrix (Table 5.6). The dataset will be enlarged by a further 50 acquisitions per stage; these have not yet been downloaded or processed, so every number in this report refers to the current 165-acquisition cohort.

- **Comment 2** — *Need more clarity on the Meta-Bayesian framework.*

Response — “Meta” and “Bayesian” are two separate parts that solve two separate problems. Meta-learning treats each biomarker as a task and trains the shared encoder to give one biomarker-invariant representation. The Bayesian ordinal head turns that representation into a distribution over the six ordered stages with an uncertainty measure. Section 4.5.1 gives the equations and a diagram. Neither part has been implemented yet.

- **Comment 3** — *Need more clarity on the comorbidity data.*

Response — The covariates are age, APOE $\epsilon 4$, MMSE, CDR-SB, GDS and cardiovascular and endocrine medical history. Section 4.5.2 lists each one with its ADNI source and current status. The cleaned master table covers all 127 subjects; APOE $\epsilon 4$ is present for 126, and the other covariates for the 107 subjects outside the LMCI group, whose source tables were not found.

3. Problem statement

The problem statement is the one fixed at Review I.

Problem. Existing computational systems analyse rs-fMRI biomarkers independently and omit clinical context, resulting in overconfident deterministic staging errors on borderline subjects for diagnostic clinicians.

The primary problem addressed by this research is the development of a unified, robust computational framework that can accurately assess continuous Alzheimer’s disease progression by synthesising heterogeneous multi-biomarker resting-state fMRI data with patient-specific clinical characteristics and comorbidity metadata. Three points frame it.

- **Context.** Dementia drives progressive cognitive decline across six ordered stages.
- **Limitation.** Current pipelines isolate complementary functional features (ALFF, ReHo, DC, FC).
- **Clinical gap.** Models ignore how patient comorbidities modify the underlying neurodegeneration.

Research question. Can a common Alzheimer’s progression representation be discovered across multiple functional biomarkers while accounting for patient comorbidity profiles and prediction uncertainty?

3.1 Input Specifications

The framework operates on a multifaceted patient vector extracted from the ADNI cohort [1, 2]:

- **Neuroimaging data:** cleaned, time-ordered BOLD signal recordings extracted from each subject’s rs-fMRI scan.
- **Anatomical representation:** region-of-interest (ROI) matrices parcellated into 246 distinct functional areas using the Brainnetome Atlas.
- **Functional biomarkers:** four complementary regional descriptors — ALFF, ReHo, FC and DC — each capturing a distinct facet of local or network-level brain activity.

- **Clinical and comorbidity covariates:** age and APOE $\epsilon 4$ genotype status; MMSE and CDR-SB; hypertension history, diabetes mellitus profiling and Geriatric Depression Scale (GDS) scores.

Diagnostic group labels are taken only from `participants.tsv`. The ADNI medical-history tables obtained carry only broad cardiovascular and endocrine categories, so the hypertension and diabetes covariates will be used under those category names (Section 6.5).

3.2 Output Specifications

The architecture maps the multi-modal inputs to a risk-calibrated output tuple:

- **6-class ordinal staging:** each subject is placed on the ordered scale $CN \rightarrow SMC \rightarrow EMCI \rightarrow MCI \rightarrow LMCI \rightarrow AD$.
- **Probabilistic confidence:** a calibrated uncertainty metric output alongside each stage, flagging boundary cases and atypical trajectories.
- **Anatomical explainability:** localised attention heatmaps showing which brain regions and macro-networks (DMN, SN, ECN) drove the prediction.

3.3 System Constraints

The framework is bounded by the following, as set at Review I:

- **High-dimensional feature spaces:** the spatiotemporal complexity of rs-fMRI requires dimensionality reduction without losing fine neural detail.
- **Scan variant irregularities:** multi-site scanner variance in ADNI demands standardised preprocessing.
- **Biomarker heterogeneity:** local synchronisation and global network centrality are different signal types and need a biomarker-invariant mapping.
- **Cross-modal alignment:** fusing high-dimensional imaging tensors with sparse scalar clinical metadata.
- **Computational burden:** the memory and training cost of cascaded global, local and network self-attention.
- **Calibration reliability:** the Bayesian ordinal output must give stable, well-calibrated uncertainty for borderline subjects.

Constraints encountered while building the data foundation

Building Modules 1–3 added concrete limits that instantiate the above.

- **No anatomical reference.** Zero of the 131 audited subjects have a T1w image, so anatomical normalisation, subject-specific tissue segmentation and contrast-to-noise estimation are unavailable.
- **No slice-timing metadata.** Slice-timing correction cannot be applied to any acquisition.
- **Scan variant irregularities.** Eight acquisitions have a TR of about 6.02 s, which breaks the fixed 0.01–0.10 Hz band; one further acquisition has TR 2.25 s, which passes the Nyquist check (0.22 Hz) and was processed normally.
- **Small effective sample.** 165 acquisitions come from only 127 distinct subjects, and the smallest group has 14. Model capacity and validation design are bounded by this, not by the acquisition count.

3.4 Evaluation Criteria

Model performance will be validated across quantitative, probabilistic and qualitative dimensions, as fixed at Review I:

- **Classification performance:** accuracy, precision, recall and macro-F1 across all six stages.
- **Discriminative performance:** macro-averaged and per-class ROC-AUC.
- **Bayesian calibration:** expected calibration error (ECE), to test borderline-subject identification.
- **Explainability:** regional correlation of the learned attention vectors with established neurodegeneration atlases.
- **Benchmark comparison:** against independent single-modality baselines and standard clinical prediction architectures.

The criteria for the data stage reported here are: complete 246/246 regional coverage for every acquisition; zero non-finite values in any derived feature; and every derived quantity reproducible by independent recomputation to within 10^{-10} . None of the model-stage criteria has been evaluated yet.

4. Methodology

The proposed system, the Comorbidity-Aware Meta-Bayesian Multi-Attention Transformer, follows the six-module architecture fixed at Review I: Data/Input Management, Preprocessing Pipeline, Functional Feature Extraction (Brainnetome 246), Proposed Deep Learning Model, Core Output Classifier and Decision Support Dashboard. Modules 2 and 3 are complete, Module 1 is complete for imaging only, and Modules 4–6 have not yet been started.

4.1 Proposed Workflow

- Step 1:** (*Module 1*) Audit the raw cohort, recover acquisitions lacking image data from DICOM, document all exclusions, and clean the clinical tables.
- Step 2:** (*Module 2*) Preprocess each acquisition into a common template space under the dataset’s metadata constraints.
- Step 3:** (*Module 3*) Parcellate into the Brainnetome-246 atlas, extract regional time series, and generate and verify ALFF, ReHo, FC and DC.
- Step 4:** (*Module 3*) Normalise within subject, without using cross-subject or label information.
- Step 5:** (*Module 4*) Encode the features with a multi-attention transformer, learn a shared biomarker-invariant representation by meta-learning, and fuse the comorbidity covariates.
- Step 6:** (*Module 5*) Classify stage with a Bayesian ordinal head that reports uncertainty.
- Step 7:** (*Module 6*) Present the stage, probabilities and attention maps in the decision-support dashboard.

4.2 Deviations from the Review I Plan

Auditing the data changed four specifications made at Review I. Each change was forced by what the dataset contains, and none was made for convenience (Table 4.1).

Table 4.1: Deviations from the Review I plan and their evidence.

Review I specified	Implemented	Evidence and consequence
Slice-timing correction (Module 2, §5.2.2)	Step omitted; recorded as NOT_PERFORMED in every provenance file	No SliceTiming field in any of 175 JSON sidecars; DICOM headers from 17 protocol signatures, including private vendor tags, give volume-level timing only. Outcome SLICE_TIMING_NOT_RECOVERABLE. The step was omitted, not assumed away, and no timing was inferred.
Co-registration to structural scans (Module 2, §5.2.2)	Direct EPI-to-MNI registration (ANTsPy SyN) with population-template tissue priors	Zero of 131 subjects have a T1w image. Subject segmentation is therefore impossible; CNR is not computable for any acquisition, and the CSF regressor failed its reliability gate in all 165 (Section 5.2).
246×4 feature matrix of ALFF, ReHo, DC and FC (Module 3, §5.2.3)	All four biomarkers kept: 246×3 (mALFF, mReHo, DC _z) plus the full 246×246 Fisher- z FC matrix	The fourth column was the per-region FC summary (FC strength), which equals DC/245 exactly, so it duplicated the DC column (Section 5.3.4). FC itself is not dropped; it moves to its full matrix form.
Cross-site scanner variance normalisation via FS-FAST (Scope, §1.4)	Not performed; only within-subject normalisation was applied	FS-FAST was used for motion correction. No site or scanner harmonisation step exists in the pipeline, so site and protocol effects (matrix size, TR) are recorded but not corrected. This is listed as a limitation.

4.3 Architectural Design for Proposed System

Figure 4.1 shows the six-module architecture of Review I, coloured by status at Review II. Data flows strictly left to right through Modules 1–5, with the dashboard (Module 6) reading the outputs of the others. Each stage reads only the outputs of the stage before it and writes to its own directory, so no stage can modify its own input. This separation is what makes each stage independently auditable.

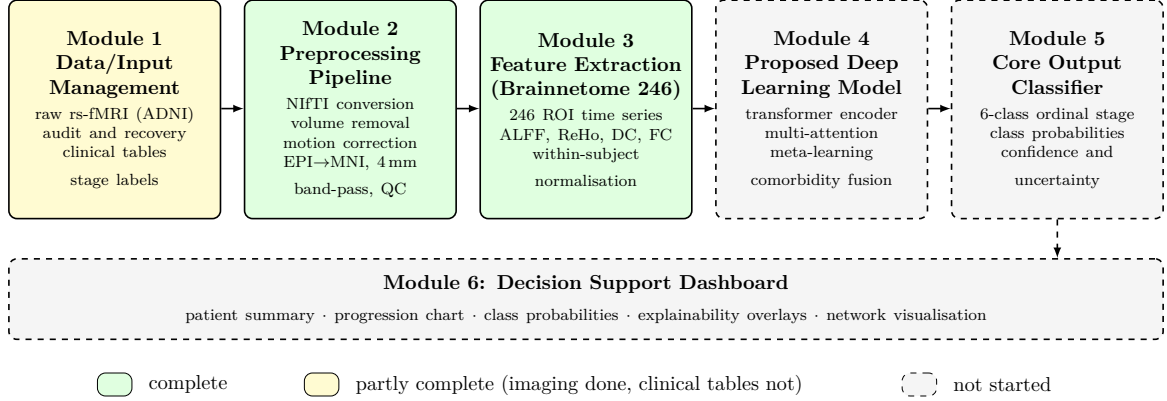


Figure 4.1: Six-module architecture of the proposed system (Review I design) with status at Review II.

4.4 Modules

Each module is described by its purpose, inputs, processing, outputs and dependencies, using the module names fixed at Review I.

4.4.1 Module 1: Data/Input Management

Purpose. Gather, structure and sanitise the raw inputs, and establish a trustworthy cohort with an explicit record of what the source data does and does not contain. **Inputs.** Raw ADNI rs-fMRI scans (DICOM and Nifti) and the clinical tables. **Processing.** Inventory every acquisition; convert and recover acquisitions lacking image data from DICOM; detect duplicate and truncated acquisitions by checksum and header inspection; audit slice-timing recoverability; organise into a BIDS-like layout with group membership taken only from `participants.tsv`; sort the clinical tables (CDR, MMSE, GDS, medical history, age, APOE $\epsilon 4$). **Outputs.** Organised cohort, exclusion list and audit reports; clinical tables bound to subjects. **Dependencies.** None. **Status.** Complete: the imaging cohort is audited and the clinical sources are cleaned into a 127-subject master table (Section 4.5.2); covariates other than APOE $\epsilon 4$ exist for 107 subjects because no LMCI source table was found.

4.4.2 Module 2: Preprocessing Pipeline

Purpose. Neutralise cross-site and acquisition variance and isolate the low-frequency BOLD signal, bringing every acquisition into a common template space. **Inputs.** Module 1 cohort. **Processing.** The eleven steps of Table 5.1. **Outputs.** One preprocessed 4D BOLD volume, brain mask, motion parameter file, nuisance design matrix, provenance record and QC report per acquisition. **Dependencies.** Module 1. **Status.** Complete for 165 of 173 eligible acquisitions.

4.4.3 Module 3: Functional Feature Extraction (Brainnetome 246)

Purpose. Translate the BOLD signal into a fixed 246-region representation and four functional biomarkers. **Inputs.** Preprocessed BOLD from Module 2; the Brainnetome atlas. **Processing.** Parcellation mapping onto each subject’s own grid with nearest-neighbour interpolation only; regional time-series extraction; voxel-wise ALFF and ReHo aggregated to regions; ROI-level FC and DC (Table 5.2); independent recomputation checks; and, as an addition to the Review I design, leakage-free within-subject normalisation (Table 5.3). **Outputs.** Per acquisition: the 135×246 ROI time series, ALFF, ReHo and DC vectors, the 246×246 FC matrix and, after normalisation, ALFF, ReHo and DC as a 246×3 regional matrix and FC as the full Fisher- z matrix. All four Review I biomarkers are kept; Review I assembled them into a 246×4 matrix, whose FC column is covered in Section 5.3.4. **Dependencies.** Module 2. **Status.** Complete for 165 of 165 acquisitions.

4.4.4 Module 4: Proposed Deep Learning Model

Purpose. Encode multi-scale functional dependencies and distil shared, biomarker-invariant disease representations. **Inputs.** Module 3 features and the clinical covariates. **Processing.** Token encoding of the region features with a Transformer; global, local and network self-attention [13]; latent-representation learning; a meta-learning loop across biomarker tasks [14]; and multimodal fusion into $H = [Z' \parallel C]$. Subject-grouped stratified cross-validation, with cross-subject scaling fitted on training folds only, is part of this module’s evaluation protocol. **Outputs.** Fused latent vector. **Dependencies.** Module 3 and the cleaned clinical tables (ready). **Status.** Not started.

4.4.5 Module 5: Core Output Classifier

Purpose. Map the fused vector to a calibrated, uncertainty-aware ordinal stage. **Inputs.** Fused vector from Module 4. **Processing.** An ordinal multi-layer perceptron head over $CN \rightarrow SMC \rightarrow EMCI \rightarrow MCI \rightarrow LMCI \rightarrow AD$ with a cumulative-link formulation [16],

$$P(y \leq k | H) = \sigma(\theta_k - g(H)), \quad \theta_0 < \theta_1 < \dots < \theta_4, \quad (4.1)$$

with Bayesian layers [15] for predictive uncertainty and flagging of borderline subjects. **Outputs.** Stage $\hat{y}$ with class probabilities and a confidence score. **Dependencies.** Module 4. **Status.** Not started.

4.4.6 Module 6: Decision Support Dashboard

Purpose. Present the results to a clinician as analytical decision support only, not as a replacement for professional evaluation. **Inputs.** Stage labels, probabilities, attention indices and patient metadata from Modules 1, 3, 4 and 5. **Processing.** Patient summary, progression chart, class probabilities, explainability overlays and network (DMN, SN, ECN) visualisation. **Outputs.** Interactive dashboard. **Dependencies.** Modules 1 and 5. **Status.** Not started.

4.5 Clarifications Requested at Review I

4.5.1 The Meta-Bayesian Framework

The Review I panel asked for more clarity on this framework. “Meta-Bayesian” combines two independent ideas (Table 4.2). Both belong to Modules 4 and 5, and neither has been implemented; what follows is the design.

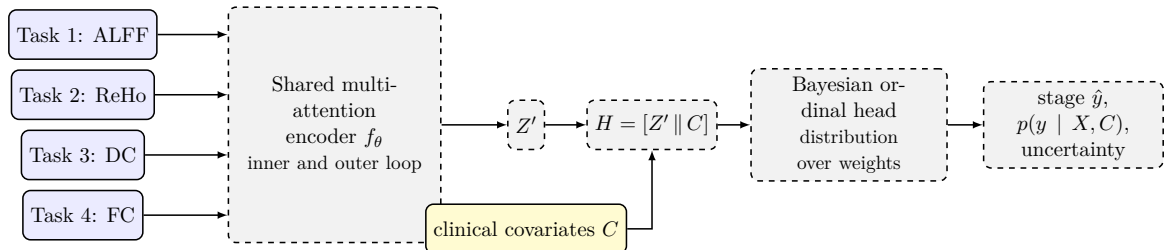


Figure 4.2: The Meta-Bayesian pipeline (design, not implemented). The four biomarkers act as meta-learning tasks; the clinical covariates are fused after the shared representation is learnt.

Meta-learning across biomarkers. Let $b \in \{ALFF, ReHo, DC, FC\}$ index

Table 4.2: The two parts of the Meta-Bayesian framework.

Part	Problem it addresses	Mechanism and output
Meta-learning (Module 4)	Each biomarker is a different, noisy view of the same disease, and a model trained on one generalises poorly.	Each biomarker is a task. A learn-to-learn loop trains the shared encoder so that it adapts quickly to any of them. Output: a biomarker-invariant representation Z' .
Bayesian ordinal head (Module 5)	Stages are ordered, and borderline subjects need an honest confidence, not a single hard label.	A cumulative-link output over the six ordered stages, with a distribution over the classifier weights. Output: stage probabilities and an uncertainty measure.

the tasks, and let f_θ be the shared encoder. For each task a *support* set adapts the encoder and a *query* set measures how well the adapted encoder works, following model-agnostic meta-learning [14]:

$$\theta'_b = \theta - \alpha \nabla_\theta \mathcal{L}_b^{\text{sup}}(\theta), \quad \theta \leftarrow \theta - \beta \nabla_\theta \sum_b \mathcal{L}_b^{\text{qry}}(\theta'_b). \quad (4.2)$$

The outer update rewards parameters θ from which a few steps work well on *every* biomarker, which is what makes $Z' = f_\theta(X)$ biomarker-invariant. Support and query sets are drawn from different subjects, and all of them come from the training folds only. The tasks here are biomarkers, not sites or subjects.

Bayesian ordinal head. For the fused vector H the head scores $g_w(H)$ and uses ordered thresholds $\theta_0 < \dots < \theta_4$ (the cumulative-link form [16]):

$$P(y \leq k \mid H, w) = \sigma(\theta_k - g_w(H)), \quad k = 0, \dots, 4. \quad (4.3)$$

The weights have a prior and an approximate posterior $q(w)$, obtained for example by variational inference or Monte Carlo dropout [15]. The predictive probability averages over S weight samples $w_s \sim q(w)$:

$$\bar{p}_k = \frac{1}{S} \sum_{s=1}^S \left[P(y \leq k \mid H, w_s) - P(y \leq k-1 \mid H, w_s) \right], \quad (4.4)$$

with $P(y \leq -1) = 0$ and $P(y \leq 5) = 1$. Uncertainty is the predictive entropy $U = -\sum_k \bar{p}_k \log \bar{p}_k$. A subject is flagged as borderline when two adjacent stages both carry substantial probability.

Status and open choices. Nothing here has been trained. First-order or full meta-gradients, variational inference or Monte Carlo dropout, and the size of S are not yet fixed. The model will be evaluated with the Review I metrics, including

expected calibration error.

4.5.2 The Comorbidity Data

The Review I panel asked for more clarity on the comorbidity data. Table 4.3 lists every covariate fixed at Review I with its ADNI source and coverage in the cleaned master table (127 subjects, 90 columns).

Table 4.3: Clinical and comorbidity covariates (Review I design) and their coverage in the cleaned master table (127 subjects).

Covariate	ADNI source	Type	Coverage
Age	image-collection tables	years	107 of 127; none for LMCI
APOE $\epsilon 4$	APOE genotyping table	$\epsilon 4$ allele count, carrier	126 of 127; counted from the genotype string
MMSE	MMSE table	score 0–30	107 of 127; none for LMCI
CDR-SB	CDR table	score 0–18	107 of 127; none for LMCI
GDS	Geriatric Depression Scale table	score 0–15	107 of 127; none for LMCI
Cardiovascular history	medical history, MH4CARD	binary	107 of 127; broad category that includes hypertension
Endocrine/metabolic history	medical history, MH9ENDO	binary	107 of 127; broad category that includes diabetes

How the table was cleaned. One script (`scripts/clinical/clean_clinical_data.py`) reads 37 ADNI clinical tables without modifying them. Each CDR, MMSE and GDS score comes from the assessment closest in date to the subject’s earliest MRI (median gap 0 days). Missing-value codes become missing, nothing is imputed, and 116 conflicting duplicate values are left for manual review.

How the covariates enter the model. They form a vector C , standardised with training-fold statistics only and concatenated as $H = [Z' \parallel C]$ (Review I design).

Missing data. No LMCI table for age, sex, MMSE, CDR, GDS or medical history was found, so the 20 LMCI subjects lack all of them (their APOE $\epsilon 4$ comes from a separate genotyping file). Because the gap follows group lines, a missingness indicator would proxy the LMCI label and must not be a feature, and nothing will be imputed for a whole missing group. Only an imaging-only analysis covers all six stages.

A circularity risk. ADNI assigns diagnostic categories partly from MMSE and CDR, so a model given MMSE and CDR-SB can read its label from the definition of that label. Results will be reported with and without these two scores; the imaging claim rests on the version without.

5. Implementation

All processing ran under WSL2 (Ubuntu) on Windows 11 with Python 3.12.3, NumPy 2.3.5, SciPy 1.15.3, NiBabel 5.4.2, Nilearn 0.14.1, ANTsPy 0.6.3, TemplateFlow 25.1.2 and FreeSurfer 7.4.1. Every long-running job used a single worker, with all thread-count environment variables pinned to one.

5.1 Module 1: Data/Input Management

The initial audit covered 131 subjects and found 72 valid NIfTI files against 103 records that had JSON metadata but no image data at the audited location. Of these, 29 were recovered by converting raw DICOM archives (9 in the MCI group and 20 in the SMC group, the latter representing 62.5% of that group’s final data), with no conversion failures; the remaining records were reconciled against NIfTI files found by the later per-group mapping audits. A dataset-wide checksum audit of 175 image files found no orphans and exactly one hash collision, a subject exported twice under different series numbers; the duplicate was excluded and the original retained. A second acquisition was excluded for containing only 7 volumes where all others contain 140, a truncation traced to the source export rather than to any processing step. Both exclusions are recorded in a canonical exclusion list that every batch script consults. An early working note recorded six AD acquisitions as unrecoverable; re-checked against the source data it was stale, and all 25 AD acquisitions were present, valid and processed.

The slice-timing audit (evidence in Table 4.1) examined all 175 JSON sidecars and DICOM headers from all 17 protocol signatures, including private vendor groups. `AcquisitionTime` and `TriggerTime` are identical across the slices of a volume, so timing is known only per volume. A Philips private tag gives a per-slice *spatial* index, but no tag states the temporal order, and the spatial index was deliberately not used to infer it. The conclusion was recorded as `SLICE_TIMING_NOT_RECOVERABLE`, and no timing value was inferred or fabricated.

Module 1 is complete: a single audited script reads the ADNI clinical tables without modifying them and writes a 127-subject master table (Section 4.5.2). Covariates other than APOE ϵ 4 exist for 107 subjects because no LMCI source table was found. A further 50 acquisitions per stage are planned; they have not yet been downloaded or processed.

5.2 Module 2: Preprocessing Pipeline

Table 5.1: Implemented preprocessing pipeline.

Step	Implementation
Input validation	4D shape, all values finite, non-degenerate affine, $TR > 0$, non-empty image
Volume removal	First 5 volumes discarded for signal equilibration
Slice timing	Not performed — not recoverable for this dataset
Motion correction	FreeSurfer [11] FS-FAST <code>mc-afni2</code> (AFNI [12] <code>3dvolreg</code>), rigid 6-DOF, reference = first retained frame; native <code>.mcdat</code> preserved
Normalisation	Direct EPI-to-MNI152Nlin6Asym via ANTsPy symmetric diffeomorphic registration (<code>SyN</code>) [10]; no T1w exists, so the skull-stripped temporal mean BOLD is registered to the template and the 4D series warped in a single interpolation
Resampling	4 mm isotropic, continuous interpolation
Smoothing	6 mm FWHM Gaussian
Detrending	Per-voxel linear OLS; intercept and linear term removed, temporal mean restored; no quadratic term
Nuisance regression	27 regressors in every acquisition: intercept, Friston-24 [9], white-matter signal from a population template prior, and global signal. A CSF candidate from the same priors failed its reliability gate in all 165 acquisitions
Band-pass filter	0.01–0.10 Hz, 2nd-order Butterworth, zero-phase <code>filtfilt</code> ; Nyquist validity checked per acquisition
QC	tSNR, SNR, framewise displacement [8], DVARS, spatial and temporal entropy

Two implementation points deserve emphasis. First, because no T1w image exists, white-matter and CSF candidate regressors are derived from published population tissue probability priors rather than from subject-specific segmentation; a candidate is included only if its mask retains at least 20 voxels and its correlation with the global signal is below 0.98 in magnitude. The white-matter regressor passed this gate in all 165 acquisitions and the CSF regressor failed it in all 165, so the nuisance design matrix has exactly 27 columns in every acquisition (intercept, 24 Friston terms, white matter, global signal); every provenance record states `CSF_status = NOT PERFORMED`. The ventricular CSF prior is too thin to survive at 4 mm: the corresponding 2 mm pilot mask held only 111 voxels (888 mm^3) after thresholding at 0.90 and intersection with the brain mask, which is roughly 14 voxels on the 4 mm grid, below the 20-voxel minimum. The count at the gate was not stored per acquisition, so this is the explanation consistent with the logged evidence, not a per-acquisition measurement. No substitute CSF regressor was introduced. Second, contrast-to-noise ratio is reported as not reliably computable for all 165 acquisitions, because it requires a tissue segmentation this dataset cannot provide (no T1w image exists for any of the

131 subjects); no segmentation was fabricated to produce a number.

The band-pass step validates, per acquisition, that the requested upper cutoff lies below the Nyquist frequency implied by that acquisition’s own TR. Eight acquisitions failed this check. All eight share a TR of approximately 6.02 s, roughly double the 3.0 s used elsewhere in the cohort, giving a Nyquist limit near 0.083 Hz — below the 0.10 Hz cutoff. These were refused rather than silently reprocessed with an altered band, since changing the filter for a subset would make those acquisitions non-comparable with the rest. Seven of the eight have a TR of 6.02158 s and one has 6.02353 s; all eight belong to site 002 (subject identifiers 002_S_XXXX).

5.3 Module 3: Functional Feature Extraction (Brainnetome 246)

5.3.1 Parcellation and Regional Time-Series Extraction

The Brainnetome atlas [3] is supplied at 2 mm in LAS orientation whereas the preprocessed BOLD is RAS, so the atlas is first reoriented to canonical orientation, a lossless array operation involving no interpolation, verified by per-label voxel count equality. It is then resampled onto each subject’s own BOLD grid using *nearest-neighbour interpolation only*. This is essential: atlas values are categorical region identifiers, and linear or spline interpolation would produce fractional labels with no meaning. Regional time series are extracted by direct index masking on the verified-identical grid rather than through a masker utility that might silently re-sample the labels a second time.

5.3.2 Biomarker Generation

ALFF and ReHo are computed at voxel level and only then aggregated to regions, so that local spatial information is used where the definitions require it. The 27-voxel ReHo neighbourhood is the original formulation [4] and the default of established toolboxes. Because the preprocessed BOLD is already band-pass filtered, ALFF is computed on an already band-limited spectrum, a documented consequence of the fixed pipeline. The connectivity matrix is kept in full: no threshold, binarisation or absolute value is applied, and negative correlations are preserved.

A single-acquisition pilot was validated first, and the production implementation was required to reproduce it before scaling up: ALFF, ReHo, regional time series, atlas labels and voxel counts were bit-identical, and FC and DC agreed to within 1.8×10^{-15} . Only then was the pilot’s output directory protected and the

Table 5.2: Implemented biomarker definitions.

Biomarker	Definition as implemented	Shape
ALFF	Mean FFT amplitude within 0.01–0.10 Hz of the mean-centred voxel time series, then averaged within each region	(246,)
ReHo	Kendall’s coefficient of concordance over a 27-voxel $3 \times 3 \times 3$ neighbourhood including the centre voxel, with the neighbour count reduced at the mask boundary, then averaged within each region	(246,)
FC	Pearson correlation between all region pairs; signed, unthresholded, diagonal retained	(246, 246)
DC	$DC_i = \sum_{j \neq i} FC_{ij}$; weighted, signed, unthresholded, self-connection excluded	(246,)

remaining acquisitions processed.

5.3.3 Within-Subject Normalisation

Table 5.3: Within-subject normalisation, computed from each acquisition’s own 246 regional values only.

Feature	Transform and rationale
mALFF	$ALFF_i / \overline{ALFF}$. Raw ALFF is in FFT amplitude units proportional to raw BOLD intensity and spans roughly 8.5×10^2 to 2.2×10^5 across acquisitions; the ratio removes this scaling [6]. A ratio is used rather than a z -score because ALFF is strictly positive
mReHo	$ReHo_i / \overline{ReHo}$, removing each subject’s global coherence offset
DC _z	$(DC_i - \overline{DC}) / \sigma_{DC}$ with ddof = 0 [7]. A z -score is used because DC is signed, which rules out min–max scaling
Fisher- z FC	$\text{arctanh}(\text{clip}(r, \pm 0.999999))$ on off-diagonal entries, variance stabilising the correlations. The diagonal is zeroed <i>before</i> the transform so that $\text{arctanh}(1) = \infty$ is never produced, then set to exactly zero; negative correlations stay negative

Leakage-free by construction. Every statistic in Table 5.3 is computed from that acquisition’s own 246 values; no cross-subject statistic and no diagnostic label is used, so the stage cannot carry information from a test subject into training. Cross-subject scaling is deferred to the modelling stage, where it will be fitted on training folds only. Outputs go to a separate directory tree, and the script refuses any write path inside a raw `biomarkers` directory.

5.3.4 Feature Redundancy: FC Strength and DC

Auditing our own features, rather than inheriting them from the design, exposed an exact redundancy. The pipeline stores a regional connectivity strength defined as the

Table 5.4: Verification of within-subject normalisation (165 acquisitions, 0 failures).

Check	Result	Check	Result
Acquisitions normalised; failures	165 / 165; 0	Fisher- z diagonal exactly zero	165 / 165
Non-finite inputs; outputs	0; 0	Fisher- z maximum asymmetry	5.329×10^{-15}
mean(mALFF), mean(mReHo) per acquisition	1.000000	Negative correlations kept negative	165 / 165
std(DC $_z$) per acquisition	1.000000	Entries clipped at ± 0.999999	0
mean(DC $_z$), largest magnitude	$\leq 1.6 \times 10^{-16}$	Fisher- z range	-2.2653 to 2.5924

mean of each region’s off-diagonal connectivity, while DC is the *sum* over the same 245 entries, so

$$\text{FCS}_i = \frac{1}{245} \sum_{j \neq i} \text{FC}_{ij} = \frac{\text{DC}_i}{245}.$$

Table 5.5 gives the measurements.

Table 5.5: Measured equivalence of FC_Strength and DC.

Comparison	Result
Raw outputs, all 165 acquisitions: $\max \text{DC}/245 - \text{FCS} $	6.939×10^{-18}
Raw outputs: minimum Pearson r across acquisitions	1.000000000000
Stage matrices (both within-subject z -scored): $\max \text{DC}_z - \text{FCS}_z $	6.661×10^{-16}
Stage matrices: Pearson r	1.0

Under any scale-invariant normalisation the two columns become numerically identical.

All four biomarkers are kept. The four biomarkers of Review I are ALFF, ReHo, DC and FC. Review I assembled them into a 246×4 matrix, which needs one value per region for each biomarker. ALFF, ReHo and DC are naturally one value per region. FC is naturally a region-by-region matrix, so its fourth column was a per-region summary of it, the FC-derived regional feature that this report calls FC strength. That summary is exactly $\text{DC}/245$, so the fourth column added nothing beyond the third. The corrected model input therefore carries FC in its full form instead (Table 5.6).

FC therefore moves from a collapsed per-region scalar into its full matrix form, which carries strictly more information, so none of the four biomarkers is lost. Only the redundant column goes. The raw FC strength files are retained unaltered. This is a correction derived from auditing the features, not a pipeline error. A different per-region FC summary that is not a rescaling of DC (for example, within-network connectivity) could be added later if the model needs one.

Table 5.6: Where each Review I biomarker sits in the model input.

Biomarker	Review I (246×4 column)	Model input now
ALFF	ALFF per region	mALFF, 246×1
ReHo	ReHo per region	mReHo, 246×1
DC	DC per region	DC _z , 246×1
FC	FC summary per region (FC strength = DC/245)	Fisher- <i>z</i> FC, 246×246

5.3.5 Complementarity of the Retained Biomarkers

Removing FC strength leaves three regional features. Carrying them separately is worthwhile only if they overlap little, so the overlap was measured. For every acquisition, the Pearson correlation across the 246 regions was computed between each pair of within-subject normalised features, and the squared correlation (the shared variance) was averaged over the 165 acquisitions (Table 5.7). The three retained pairs share 5.7 to 11.3% of their variance, so the features are 89 to 94% non-redundant, whereas the removed pair shares 100%. mALFF and DC_z are negatively related in all 165 acquisitions: regions with larger low-frequency amplitude tend to have lower degree centrality. Low overlap shows that the features describe different aspects of each acquisition, not that any of them is informative about disease; whether separate tasks improve staging can only be tested in the modelling stage.

Table 5.7: Overlap between biomarker pairs across the 246 regions, over all 165 acquisitions. Shared variance is the mean of the squared correlation.

Pair	Mean <i>r</i>	Shared variance	Acquisitions with $r < 0$
FC_Strength–DC (removed)	+1.000	100%	0 / 165
mALFF–mReHo	+0.291	11.3%	7 / 165
mALFF–DC _z	−0.310	11.1%	165 / 165
mReHo–DC _z	−0.161	5.7%	133 / 165

5.4 Engineering and Reproducibility

5.4.1 An Unstable Execution Environment

Early production runs failed repeatedly: the virtual machine terminated after roughly five to twelve acquisitions with no Python traceback, caused by sustained small-granularity array input/output against the Windows drive mount (Errno 5). Three changes resolved it. Each acquisition is staged on the native Linux filesystem and copied to the Windows tree only once complete, so a crash never leaves a partial output directory. Resume validation reads array shapes through memory mapping and

file headers instead of reloading every finished acquisition. A wrapper distinguishes a genuine Python error, which stops the run for inspection, from a virtual-machine failure, which restarts and resumes from the last validated checkpoint. The restart count fell from 23 to 2 for the same workload. Checkpointing re-validates the files on disk (shapes, finiteness and structural invariants), so the checkpoint record alone is never trusted.

5.4.2 Code and Reproducibility

All code is in the project repository (github.com/HarishkannaR11/FYP). The scripts below are the live pipeline; earlier native-space and pilot scripts are kept in `pipeline/` and `scripts/` for provenance only and must not be used for new work (Table 5.8).

Table 5.8: The live pipeline scripts and where their outputs go.

Stage	Script	Output
Data audit and recovery	<code>scripts/audit/</code> , <code>scripts/</code>	<code>audit/</code> reports, BIDS-style cohort
Preprocessing	<code>data_prep/</code> <code>pipeline/</code> <code>fsfast_production_4mm6mm.py</code> , run through <code>scripts/setup/</code> <code>retry_all_groups_4mm6mm.sh</code>	<code>derivatives/fsfast/</code>
Preprocessing	<code>scripts/audit/</code> <code>final_6group_audit.py</code>	<code>FINAL_6GROUP_AUDIT/</code>
audit	<code>scripts/biomarkers/</code> <code>biomarker_dataset_pipeline.py</code> , run through <code>retry_biomarker_dataset.sh</code>	<code>biomarkers/</code> per acquisition; <code>BIOMARKER_DATASET_AUDIT/</code>
Parcellation and biomarkers	<code>scripts/biomarkers/</code> <code>normalize_within_subject.py</code>	<code>derivatives/</code> <code>biomarkers_normalized/</code>
Normalisation	<code>scripts/biomarkers/</code> <code>stage_heatmaps.py</code>	<code>derivatives/STAGE_HEATMAPS/</code>
Stage heatmaps	<code>scripts/clinical/</code> <code>clean_clinical_data.py</code>	<code>derivatives/clinical_cleaned/</code>
Clinical cleaning		

Every script is re-runnable: preprocessing and biomarker generation use validated checkpoint and resume, and each preprocessed acquisition carries a provenance file with its parameters, software versions and source checksum. The report's figure panels are rebuilt by `figures/make_stage_panel.py` and `figures/make_group_fc.py`.

6. Results and Discussion

6.1 Cohort Composition

Table 6.1: Cohort composition after audit, recovery and exclusion, listed in progression order.

Stage	Eligible	Preprocessed	Biomarkers	Subjects
CN (CN_Final)	29	27	27	26
SMC (SMC_Final)	32	31	31	21
EMCI	30	29	29	26
MCI	35	32	32	14
LMCI	22	21	21	20
AD	25	25	25	20
Total	173	165	165	127

Table 6.1 reports the figure that most constrains the modelling stage. The 165 acquisitions derive from only 127 distinct subjects: 96 subjects contribute one acquisition, 25 contribute two, 5 contribute three and 1 contributes four. The MCI group illustrates why this matters. By acquisition count it is the largest group at 35, but it comprises only 14 subjects, at 2.3 acquisitions per subject. Because validation splits must group by subject — otherwise the same individual appears in both training and test data — 14 is the number that bounds what can be claimed about that class. No subject appears in more than one group, so labels are subject-consistent.

6.2 Preprocessing Quality Control

Table 6.2 summarises acquisition-level quality control by stage. One acquisition is treated separately and is flagged explicitly.

Dataset-wide medians over the 164 acquisitions are tSNR 127.95, SNR 13.12, mean framewise displacement 0.317 mm and raw-intensity DVARS 15270. Contrast-to-noise ratio is not available (N/A) for all 165 acquisitions, since no T1w image exists for any of the 131 subjects. Head motion is substantial: 134 of the 164 acquisitions exceed 0.2 mm mean FD, 90 exceed 0.3 mm and 28 exceed 0.5 mm. High motion is flagged, not excluded, under the frozen processing policy, so motion will enter the modelling stage as a covariate.

Artifact acquisition (reported, not silently dropped). The audit-reported CN tSNR mean of 12064 is produced by a single acquisition, `CN_Final/sub-018S4313/ses-01/run-01`, whose tSNR is 322,315 against a dataset median

Table 6.2: Preprocessing quality control by stage. All columns except the last exclude the single artifact acquisition described below ($n = 164$ overall). The raw tSNR mean as originally reported by the dataset audit, which includes it, is shown in the last column. FD is mean framewise displacement in mm; “med.” is the median.

Stage	n	tSNR mean	tSNR med.	SNR mean	FD mean	tSNR raw
CN (CN_Final)	26	131.63	132.58	12.95	0.352	12064.36
SMC (SMC_Final)	31	125.71	123.70	13.15	0.375	125.71
EMCI	29	137.49	133.16	13.76	0.355	137.49
MCI	32	137.05	139.24	12.94	0.321	137.05
LMCI	21	127.54	121.41	12.99	0.322	127.54
AD	25	114.06	118.76	13.49	0.439	114.06
All	164	129.40	127.95	13.22	0.360	2082.04 ($n = 165$)

of 128. The ratio is mean over standard deviation per voxel, so near-zero-variance voxels inflate it; the project methodology record attributes the value to this effect. The same acquisition has mean FD 1.079 mm (among the four highest of 165) and an anomalously low spatial entropy of 4.21 bits against a cohort mean of 7.18 for the other 164. It is excluded from the tSNR, SNR, FD and DVARS summaries in Table 6.2 and from the medians above, and the raw value is kept visible in the last column. With it removed, the CN tSNR mean is 131.63 (median 132.58), in line with the other stages. The acquisition itself remains among the 165 preprocessed acquisitions and in every downstream count; how it is handled in modelling has not yet been decided.

6.3 Biomarker Quality

The result in Table 6.3 that carries the most evidential weight is the independent recomputation of degree centrality. For every acquisition, DC was recomputed from the stored connectivity matrix in a separate process that read only saved files, and compared against the stored vector; the maximum difference across all 165 acquisitions was exactly zero. This verifies not merely that the files are well formed but that the stored values are the ones the documented definition produces.

Complete 246/246 regional coverage in every acquisition follows from the pipeline design: because all acquisitions are registered to the same template-derived 4 mm grid, the aligned atlas is identical across the cohort. This was nonetheless verified per acquisition rather than assumed.

Normalisation was applied to all 165 acquisitions with no failures (Table 5.4). Mean mALFF and mean mReHo equal 1.000000 for every acquisition; DC_z has mean below 1.6×10^{-16} and standard deviation 1.000000; the Fisher- z connectivity matrix is symmetric to 5.3×10^{-15} with a diagonal of exactly zero in all 165 cases. No off-

Table 6.3: Dataset-level quality control over all 165 processed acquisitions.

Check	Result
Acquisitions with complete biomarker outputs	165 / 165
Processing failures at the biomarker stage	0
Brainnetome coverage 246/246 regions	165 / 165
Regions with zero voxels (any acquisition)	0
Smallest region, across the whole dataset	10 voxels
Regional time-series matrices of shape (135, 246)	165 / 165
Non-finite values in regional time series	0
Regions with zero temporal variance	0
ALFF: non-finite or negative values	0
ReHo: values outside the valid $[0, 1]$ range	0
ReHo range across dataset	0.2352–0.8231
FC: non-finite values	0
FC maximum diagonal error	2.2×10^{-16}
FC maximum asymmetry	2.2×10^{-16}
DC recomputed independently from stored FC, max. difference	0.0×10^0
Acquisitions passing all cross-file consistency checks	165 / 165

diagonal correlation anywhere in the dataset exceeded the ± 0.999999 clipping bound, so that safeguard never engaged — confirming no region pair is numerically perfectly correlated.

Connectivity distribution. Region sizes on the 4 mm grid range from 10 to 186 voxels (median 68). Each acquisition gives 30,135 unique region pairs, of which 50.1% are negative on average over the 165 acquisitions (45.6 to 53.1% in any single one). This is expected after global signal regression, part of the 27-regressor design, which shifts correlations toward zero, so negative values should not be read as anticorrelation between networks. The matrices are kept signed and unthresholded.

Reproducibility between sessions. Thirty-one subjects have repeat acquisitions (69 in all; 45 of the 46 same-subject pairs come from different sessions). Using no stage label, each acquisition was matched to the other acquisition whose 246-region vector correlates most strongly with it (Table 6.4). For mALFF the match belongs to the same subject in 63 of 69 cases (91.3%), against 3.2% expected by chance for 31 subjects; the rate is 76.8% for mReHo and 43.5% for DC_z .

Positive control. Left–right homologous regions are among the most strongly connected pairs in resting-state fMRI. Their mean Fisher- z connectivity is 0.63 against 0.01 for the other 30,012 pairs, and exceeds that of the other pairs in 164 of 165 acquisitions. The exception, EMCI sub-012S4849/ses-01/run-01, has a mean homologous connectivity of -0.003 , a tSNR of 87 (lowest 7% of the cohort) and the widest spread of connectivity values of any acquisition; it is not excluded and should be inspected before modelling. These checks show that the features are stable within a subject and

Table 6.4: Between-session reproducibility (31 subjects with repeat acquisitions, 69 acquisitions). Correlations across the 246 regions: mean over same-subject pairs and over 3000 random pairs from different subjects. Identification: the most similar other acquisition belongs to the same subject (chance = $1/31 = 3.2\%$).

Feature	Same subject	Different subjects	Identification
mALFF	0.83	0.46	91.3% (63 / 69)
mReHo	0.64	0.37	76.8% (53 / 69)
DC _z	0.58	0.41	43.5% (30 / 69)

reproduce a known anatomical pattern, not that they separate stages. A subject’s repeat acquisitions usually share a site and scanner, so part of the reproducibility may be scanner-related.

Granularity. The same matching test on mALFF asks whether 246 regions keep information that coarser versions lose. Merging consecutive atlas labels lowers the identification rate at every step: 91.3% with 246 regions, 87.0% with 123 (each left–right pair merged), then 84.1, 72.5 and 50.7% with 82, 41 and 6. Returns flatten, however: random subsets of 100 and 200 regions give 88.3 and 90.6%. Two regions share only 4.0% of their variance on average, and 79 principal components are needed for 95% of it. This does not show that 246 regions are optimal for staging, which only the trained model can test.

6.4 Stage-wise Group Patterns: A Negative Univariate Result

We tested whether a univariate, per-region group-mean signal distinguishes the six progression stages. Stage-wise matrices of size 246×6 were generated for mALFF, mReHo, DC_z and FC_Strength with subject-level aggregation: acquisitions were averaged within subject first and subjects then averaged within stage, so each of the 127 subjects counts once although there are 165 acquisitions. The subject counts are CN 26, SMC 21, EMCI 26, MCI 14, LMCI 20 and AD 20. Each region was then z-scored across the six stages so that colour shows how a region changes across stages and not how large it is (Figure 6.1). The absolute-value versions (`derivatives/STAGE_HEATMAPS/*_absolute.png`) are available but are not shown, because they are dominated by anatomical differences between regions.

The result (Table 6.5) is negative and is reported as such, in four steps.

1. We tested for a univariate, per-region group-mean progression signal.
2. It is not detectable. Only 1, 0 and 1 of 246 regions are monotonic from CN to

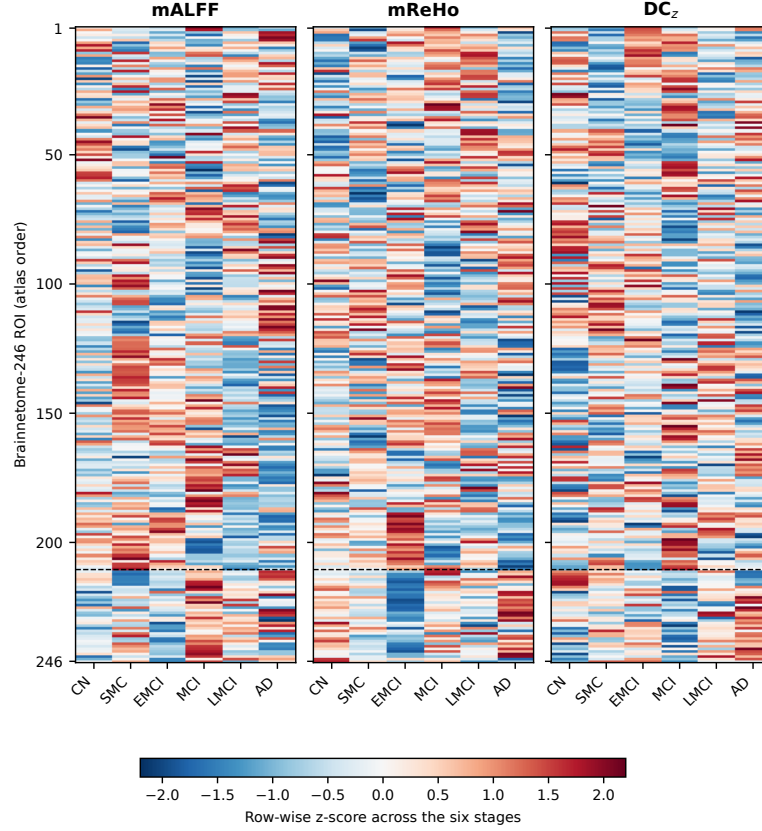


Figure 6.1: Stage-wise regional biomarker patterns across the six-stage progression (CN, SMC, EMCI, MCI, LMCI, AD) for 127 subjects; from left to right mALFF, mReHo and DC_z . Rows are the 246 Brainnetome ROIs in atlas order (cortical 1–210 above the dashed line, subcortical 211–246 below); columns are progression stages. Each ROI is z -scored independently across the six stages, so colour encodes how a region changes across stages rather than its absolute magnitude; one colour bar applies to all three panels. Acquisitions were averaged within subject before averaging across subjects, so each of the 127 subjects contributes once. These are descriptive, exploratory visualisations; they do not demonstrate disease progression and no statistical testing was performed.

Table 6.5: Monotonic trends in the stage means. A region counts as monotonic if its six stage means increase, or decrease, strictly from CN to AD. The expectation by chance is $246 \times 2/6! \approx 0.7$, the combinatorial value when the six stage means of a region are exchangeable; it is not a hypothesis test. The last column is the median over regions of the standard deviation of the six stage means, divided by the standard deviation across regions of the per-region mean (population standard deviations).

Biomarker	ROIs with monotonic CN→AD trend	Expected by chance	Stage SD / between-ROI SD
mALFF	1 / 246 (decreasing)	≈ 0.7	0.18
mReHo	0 / 246	≈ 0.7	0.24
DC_z	1 / 246 (increasing)	≈ 0.7	0.23

AD, which is no more than the roughly 0.7 that chance alone gives, and the variation across stages is about 4–5 times smaller than the anatomical variation between regions (ratios 0.18–0.24).

3. This is informative, not a failure of the pipeline. Per-region stage means discard the entire 246×246 connectivity structure, and they can only reveal effects large enough to survive averaging over 14–26 subjects per stage. The result does not show that the biomarkers carry no stage information; it shows that group means of single regions do not expose it.
4. The multivariate architecture is therefore motivated and not assumed: the transformer sees each subject individually and all 246 regions jointly, with connectivity intact.

No heatmap in this report is evidence of progression, and the pipeline has not been tested on any classification task.

Functional connectivity itself. FC is a 246×246 matrix, not a vector, so its stage-wise view is the group-mean Fisher- z connectivity matrix of each stage (Figure 6.2), averaged within subject first so that each of the 127 subjects counts once. The per-region summary of FC, FC strength, is shown beside DC_z in Figure 6.3. All six stages show the same broad structure: strong positive coupling within lobes along the diagonal, a distinct subcortical block, and weak or negative coupling between several lobes. The figure is descriptive only; no statistical test was run on it, and a visible difference between two group means would not show that individuals can be told apart.

Figure 6.3 places the DC_z and FC_Strength heatmaps side by side, which turns the redundancy of Section 5.3.4 into a visual proof.

6.5 Work Not Yet Done

To avoid any ambiguity about the scope of this report, the following are *not* done.

- **LMCI covariates are missing, and comorbidity patterns are not analysed.** No LMCI table for age, sex, MMSE, CDR, GDS or medical history was found, leaving 20 subjects with APOE $\epsilon 4$ only, and 116 conflicting duplicate values in the clinical sources await manual review. Cardiovascular and endocrine history are available only as the broad ADNI categories MH4CARD and MH9ENDO, and no analysis of comorbidity patterns or fusion with the imaging features has been done.

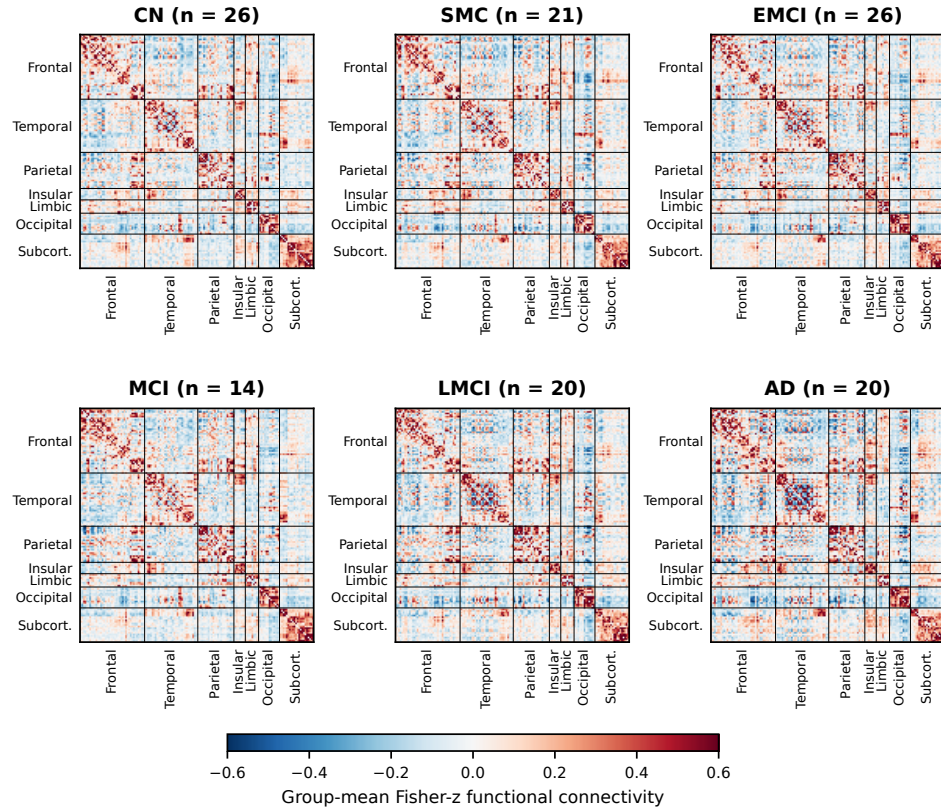


Figure 6.2: Group-mean Fisher-z functional connectivity (246×246) for each of the six stages, with the number of subjects per stage in each panel title. Regions are in Brain-netome atlas order, grouped by lobe (frontal 1–68, temporal 69–124, parietal 125–162, insular 163–174, limbic 175–188, occipital 189–210, subcortical 211–246). Acquisitions were averaged within subject before averaging across subjects. One colour bar applies to all panels. Descriptive and exploratory only.



Figure 6.3: DC_z (left) and $FC_Strength$ (right), rendered independently from separate stored files. The heatmap panels are identical, because $FC_Strength = DC/245$ exactly (maximum difference 6.939×10^{-18} across all 165 acquisitions, Pearson $r = 1.00000000000000$); the plotted row-wise z values differ by at most 6.4×10^{-15} . The right-hand file is wider only because its title carries a “derived” tag. $FC_Strength$ therefore carries no information independent of DC .

- **Motion, site and one flagged acquisition are not handled.** Mean frame-wise displacement and site are measured but not yet used as covariates or for harmonisation, and the EMCI acquisition without homologous connectivity has not been inspected.
- **The dataset expansion has not started.** The planned 50 further acquisitions per stage have not been downloaded or processed; all counts in this report refer to the current 165 acquisitions.
- **Modules 4–6 are not started:** Module 4 (multi-attention transformer, meta-learning, comorbidity fusion), Module 5 (Bayesian ordinal classifier) and Module 6 (decision-support dashboard). No accuracy, AUC or other performance figure exists, and none is implied anywhere in this report.
- **Cross-validation is not built.** Subject-level folds and train-fold-only cross-subject scaling are still to be implemented.

6.6 Discussion

Comparison with conventional practice. The pipeline departs from a textbook rs-fMRI workflow at two points, both forced by the data. A conventional pipeline co-registers BOLD to a subject T1w image and derives tissue regressors from that subject’s own segmentation; neither is possible here, so EPI-to-template registration is

direct and tissue regressors come from population priors. This is a genuine limitation: direct EPI registration is less precise, and its quality cannot be benchmarked against a T1w-mediated alternative with this dataset. Likewise, slice-timing correction is unavailable (Chapter 5), so the step was omitted and documented instead of assuming an acquisition order.

On scale and feature redundancy. Two observations about the features will shape the modelling stage more than any preprocessing choice. First, raw ALFF varies by more than two orders of magnitude across acquisitions, because it is expressed in units proportional to raw BOLD intensity, whereas ReHo, being bounded, varies only narrowly; supplying both unnormalised would let scanner and protocol scaling dominate any biological signal, which the within-subject normalisation of Module 3 addresses. Second, the per-region summary of FC, the fourth column of the Review I matrix, is an exact rescaling of DC; all four biomarkers are still used, as three regional features and FC in full. Both were found by inspecting the data rather than inherited from the design, which shows the value of auditing features before modelling.

On complementarity and reproducibility. The retained biomarkers overlap little (Section 5.3.5) and are stable between sessions (Section 6.3), which supports one task per biomarker in the meta-learning design. Neither check uses a stage label, so neither shows that the features separate stages.

Limitations. Beyond the absence of T1w and slice-timing data, five limitations bear on the conclusions that can eventually be drawn. The effective sample is 127 subjects across six ordered classes with a smallest class of 14, which bounds model capacity and makes a single held-out test split uninformative; repeated subject-grouped cross-validation is required. No cross-site or cross-scanner harmonisation was applied (Table 4.1), and sites are spread unevenly across stages (Table 6.6): one site supplies 40% of the AD acquisitions and two supply 68%, whereas no site supplies more than 19% of the CN acquisitions, so any CN–AD contrast is partly a site contrast. Motion also survives the 27 regressors: across 164 acquisitions (artifact acquisition excluded), 129 of 246 regions in mALFF, 131 in mReHo and 39 in DC_z correlate with mean framewise displacement at $p < 0.05$ (uncorrected; about 12 expected by chance; repeat acquisitions of one subject are not independent). A site covariate or harmonisation, and mean framewise displacement as a covariate, are therefore needed before modelling. Eight acquisitions are permanently excluded on account of the 6.02s TR subgroup unless the filter specification is revisited for them. Finally, Section 6.5 records what has not yet been done.

Mitigation: dataset expansion. The main limitation is the small effective sample, which counts subjects, not acquisitions (MCI has 32 acquisitions from only 14

Table 6.6: Site concentration by stage (site = three-digit prefix of the ADNI subject identifier); the last two columns are the share of the stage’s acquisitions from its largest and its two largest sites.

Stage	Acquisitions	Sites	Largest site	Two largest
CN	27	11	19%	37%
SMC	31	10	19%	32%
EMCI	29	7	21%	41%
MCI	32	8	31%	47%
LMCI	21	7	29%	48%
AD	25	7	40%	68%

subjects). A further 50 acquisitions per stage (about 300 in total) are planned; they help in proportion to the number of *new subjects* they add (MCI could grow from 14 to at most 64), so they will be drawn from subjects not already in the cohort wherever ADNI allows, and counts will be reported per subject. The frozen Modules 1 to 3 are reused unchanged.

Next steps. Immediate work is to construct the subject-level cohort index with fixed stratified grouped cross-validation folds so that cross-subject scaling is fitted on training data only, with mean framewise displacement and site carried alongside as covariates, and then implement the multi-attention transformer and meta-learning blocks. The stage order fixed at Review I (CN → SMC → EMCI → MCI → LMCI → AD) determines the loss structure of the ordinal model and is held fixed.

6.7 Revised Plan for the Remaining Work

Figure 6.4 revises the Review I plan for October to March. It adds the work this report has shown to be necessary (cross-validation folds and the dataset expansion) and keeps the Review I dates for the modelling, testing and documentation stages. It is a proposal and has not been agreed with the supervisor.

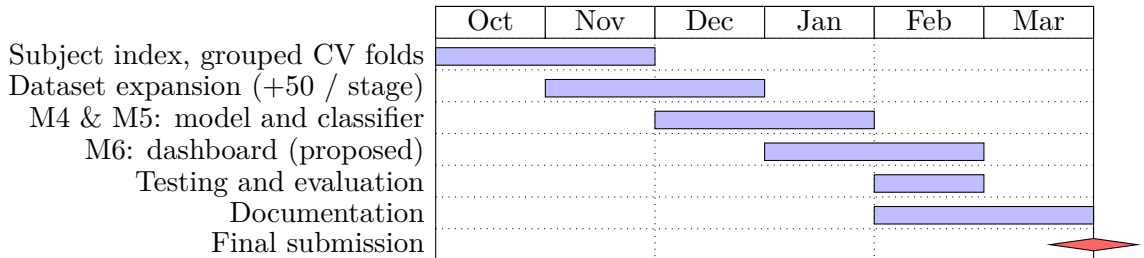


Figure 6.4: Proposed revised plan, October to March. The first two rows are new since Review I, and Module 6 had no separate row there, so its dates are a proposal; the other rows keep the Review I dates.

7. Conclusion

This report establishes the data foundation of the Comorbidity-Aware Meta-Bayesian framework. Of 175 ADNI rs-fMRI acquisitions audited, 165 from 127 subjects were preprocessed with a single frozen pipeline, parcellated into the 246 Brainnetome regions, turned into ALFF, ReHo, DC and FC biomarkers, and normalised within subject. Every derived value was verified independently, including an exact recomputation of degree centrality for all 165 acquisitions. Every limitation of the BOLD-only dataset, from the missing T1w images and slice-timing metadata to the eight scans whose 6.02s TR breaks the fixed band, was handled by documented exclusion or substitution and not by fabrication.

Four findings shaped the design. First, the per-region FC summary of Review I is exactly DC/245, so all four biomarkers are kept but enter the model as a 246×3 regional matrix and a 246×246 Fisher- z connectivity matrix. Second, the three regional biomarkers share only 6 to 11% of their variance, so each carries largely different information. Third, stage-wise group-mean heatmaps show no detectable univariate progression signal, which motivates the multivariate model of Modules 4 and 5 instead of weakening it. Fourth, the features are reproducible between sessions (91% closest-acquisition identification for mALFF, with no stage label), although residual head motion and an uneven spread of sites across stages remain to be handled in the modelling stage.

On the Review I timeline, Modules 2 and 3 are complete ahead of schedule, while Module 1 is also complete, with covariates for 107 of 127 subjects (none for LMCI). Modules 4 to 6 have not been started, and no accuracy, AUC or other performance figure exists. The next steps are to build the subject-grouped cross-validation folds, enlarge the dataset by a further 50 acquisitions per stage, and then implement the meta-learning and Bayesian ordinal model (Section 6.7).

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